

N. GREGORY

MANKIW

PRINCIPLES OF

ECONOMICS

Eighth Edition



CHAPTER

13

The Costs of Production

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You run Ford Motor Company.

- List three different costs you have.
- List three different business decisions that are affected by your costs



Look for the answers to these questions:

- What is a production function? What is marginal product? How are they related?
- What are the various costs? How are they related to each other and to output?
- How are costs different in the short run vs. the long run?
- What are “economies of scale”?



Total Revenue, Total Cost, Profit

- We assume that the firm's goal is to maximize profit.

$$\text{Profit} = \text{Total revenue} - \text{Total cost}$$

the amount a firm receives from the sale of its output

$$TR = P \times Q$$

the market value of the inputs a firm uses in production



Costs: Explicit vs. Implicit

- 'The cost of something is what you give up to get it.'
- Explicit costs
 - Require an outlay of money
 - E.g., paying wages to workers.
- Implicit costs
 - Do not require a cash outlay
 - E.g., the opportunity cost of the owner's time.
- Total cost = Explicit + Implicit costs

Explicit vs. Implicit Costs: An Example

You need \$100,000 to start your business. The interest rate is 5%.

- Case 1: borrow \$100,000
 - explicit cost = \$5000 interest on loan
- Case 2: use \$40,000 of your savings, borrow the other \$60,000
 - explicit cost = \$3000 (5%) interest on the loan
 - implicit cost = \$2000 (5%) foregone interest you could have earned on your \$40,000.

In both cases, total (exp + imp) costs are \$5000



Economic Profit vs. Accounting Profit

- Accounting profit
=total revenue minus total explicit costs
- Economic profit
=total revenue minus total costs (including explicit and implicit costs)
- Accounting profit ignores implicit costs, so it's higher than economic profit.

Active Learning 2

Economic profit vs. accounting profit

The equilibrium rent on office space has just increased by \$500/month.

Determine the effects on accounting profit and economic profit if:

- a.** you rent your office space
- b.** you own your office space

The rent on office space increases \$500/month.

a. You rent your office space.

- Explicit costs increase \$500/month. Accounting profit & economic profit each fall \$500/month.

b. You own your office space.

- Explicit costs do not change, so accounting profit does not change.
- Implicit costs increase \$500/month (opp. cost of using your space instead of renting it) so economic profit falls by \$500/month.



Production Function

- Production function
 - Relationship between
 - Quantity of inputs used to make a good
 - And the quantity of output of that good
 - Gets flatter as production rises

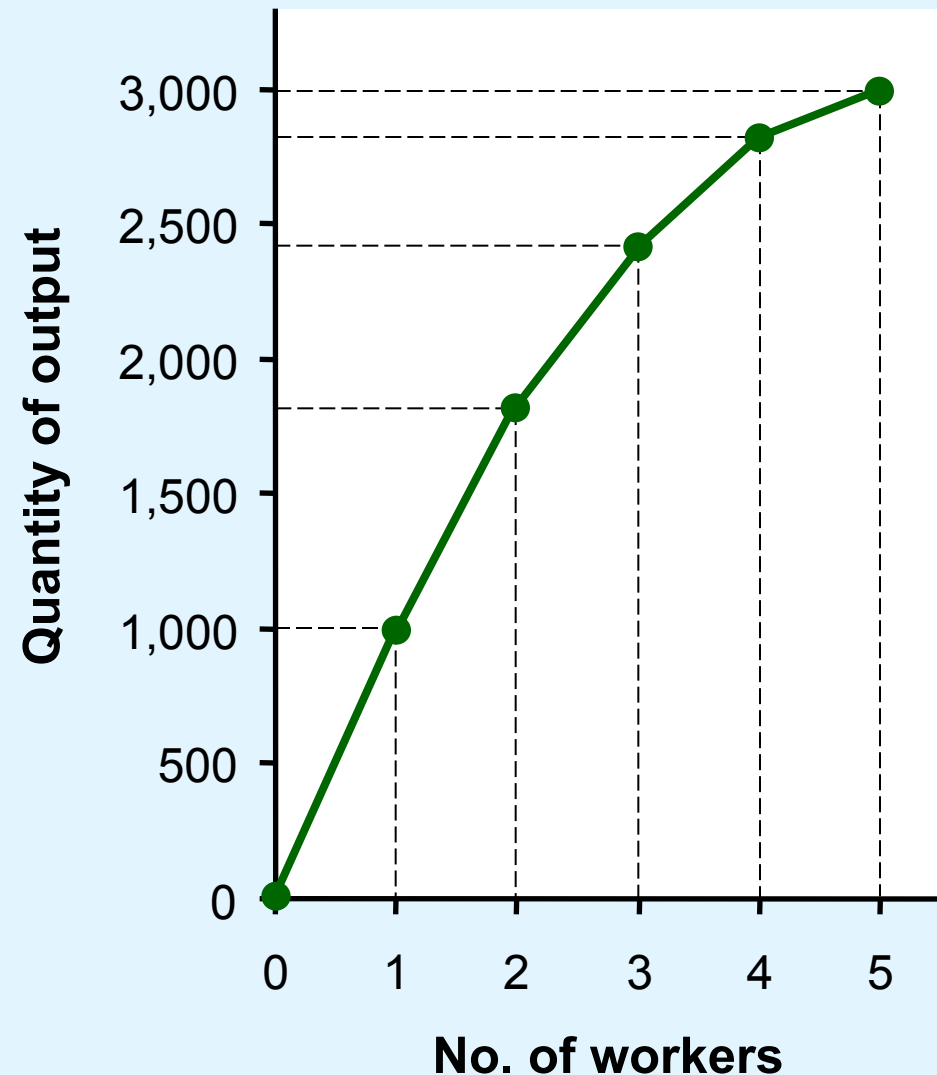
EXAMPLE 1: Farmer Jack

Example 1:

- Farmer Jack grows wheat.
- He has 5 acres of land (fixed resource).
- He can hire as many workers as he wants.
 - The quantity of output produced varies with the number of workers hired

EXAMPLE 1: Farmer Jack's Production Function

<i>L</i> (no. of workers)	<i>Q</i> (bushels of wheat)
0	0
1	1000
2	1800
3	2400
4	2800
5	3000





Marginal Product

- Marginal product
 - Increase in output that arises from an additional unit of input
 - Other inputs constant
 - Slope of the production function
- Marginal product of labor, MPL
 - $MPL = \Delta Q / \Delta L$
 - If Jack hires one more worker, his output rises by the marginal product of labor.

EXAMPLE 1: Total & Marginal Product

	<i>L</i> (no. of workers)	<i>Q</i> (bushels of wheat)		<i>MPL</i>
	0	0		
$\Delta L = 1$	1	1000	$\Delta Q = 1000$	1000
$\Delta L = 1$	2	1800	$\Delta Q = 800$	800
$\Delta L = 1$	3	2400	$\Delta Q = 600$	600
$\Delta L = 1$	4	2800	$\Delta Q = 400$	400
$\Delta L = 1$	5	3000	$\Delta Q = 200$	200

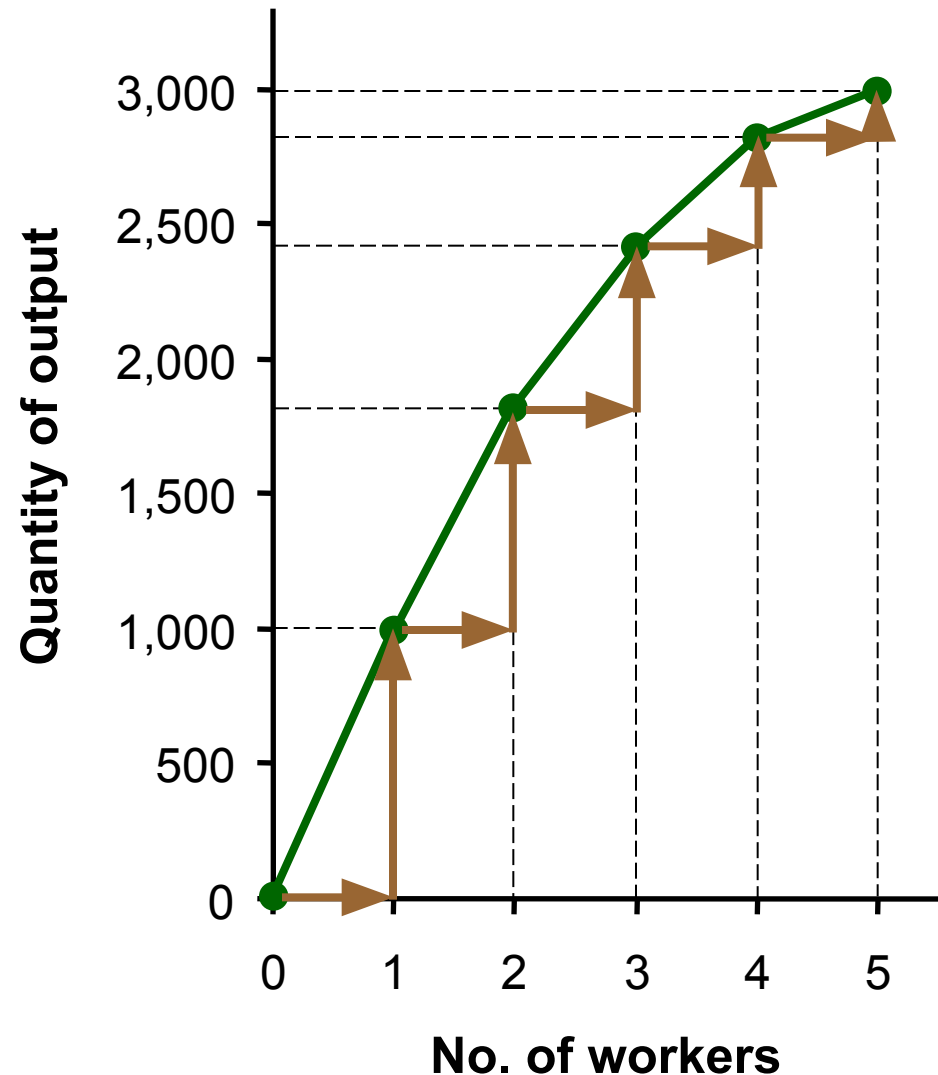


Diminishing MPL

- Diminishing marginal product
 - Marginal product of an input declines as the quantity of the input increases
 - Production function gets flatter as more inputs are being used:
 - The slope of the production function decreases

EXAMPLE 1: MPL = Slope of Prod Function

L (no. of workers)	Q (bushels of wheat)	MPL
0	0	1000
1	1000	800
2	1800	600
3	2400	400
4	2800	200
5	3000	





Why MPL Is Important

- ‘Rational people think at the margin’
- When Farmer Jack hires an extra worker
 - His costs rise by the wage he pays the worker
 - His output rises by MPL
 - Comparing them helps Jack decide whether he should hire the worker.



Why MPL Diminishes

- Farmer Jack's output rises by a smaller and smaller amount for each additional worker. Why?
 - As Jack adds workers, the average worker has less land to work with and will be less productive.
 - In general, MPL diminishes as L rises whether the fixed input is land or capital (equipment, machines, etc.).

EXAMPLE 1: Farmer Jack's Costs

Farmer Jack must pay \$1000 per month for the land, regardless of how much wheat he grows.

The market wage for a farm worker is \$2000 per month.

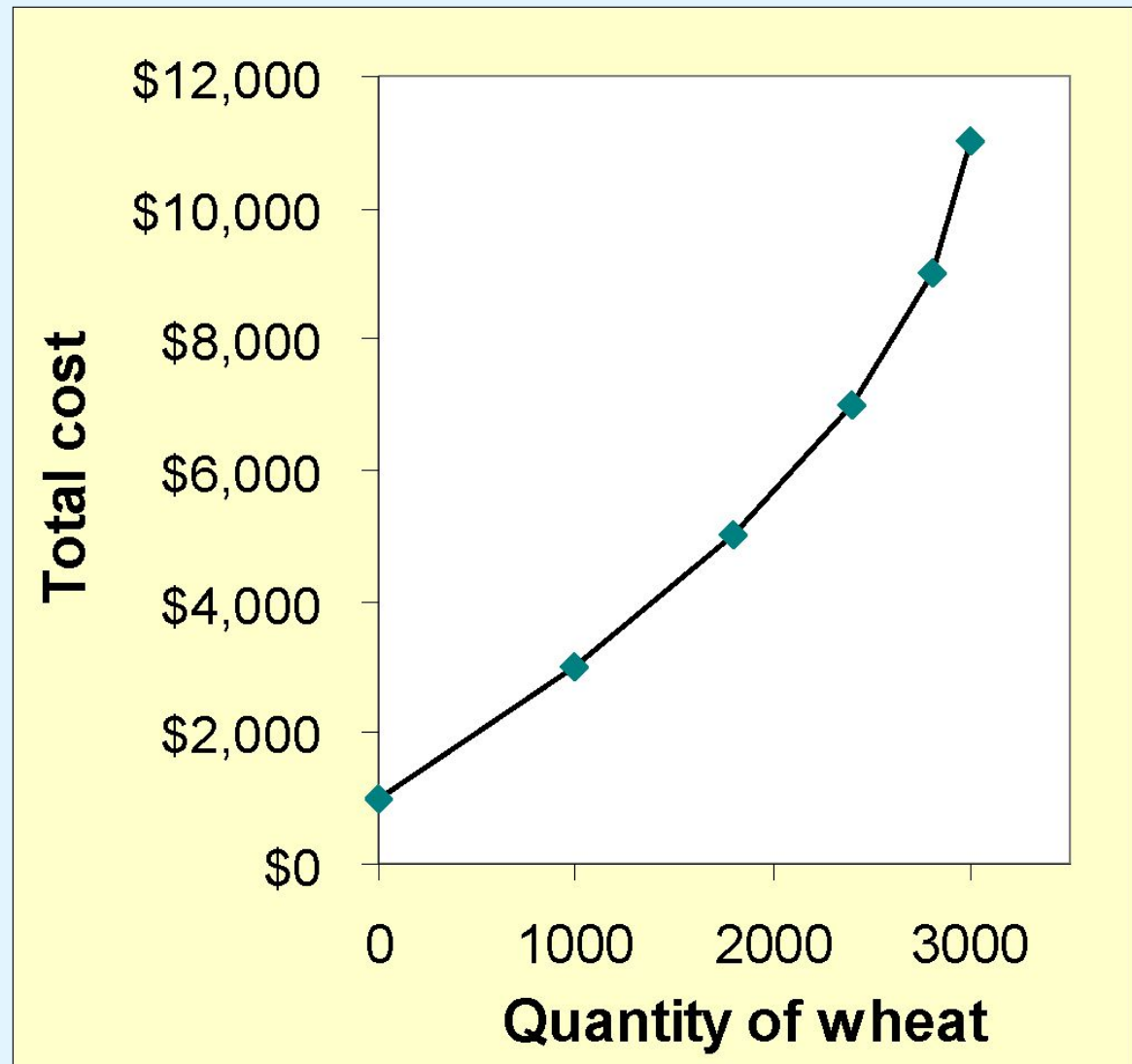
- So Farmer Jack's costs are related to how much wheat he produces....

EXAMPLE 1: Farmer Jack's Costs

<i>L</i> (no. of workers)	<i>Q</i> (bushels of wheat)	Cost of land	Cost of labor	Total cost
0	0	\$1,000	\$0	\$1,000
1	1000	\$1,000	\$2,000	\$3,000
2	1800	\$1,000	\$4,000	\$5,000
3	2400	\$1,000	\$6,000	\$7,000
4	2800	\$1,000	\$8,000	\$9,000
5	3000	\$1,000	\$10,000	\$11,000

EXAMPLE 1: Farmer Jack's Total Cost Curve

Q (bushels of wheat)	Total Cost
0	\$1,000
1000	\$3,000
1800	\$5,000
2400	\$7,000
2800	\$9,000
3000	\$11,000





Marginal Cost

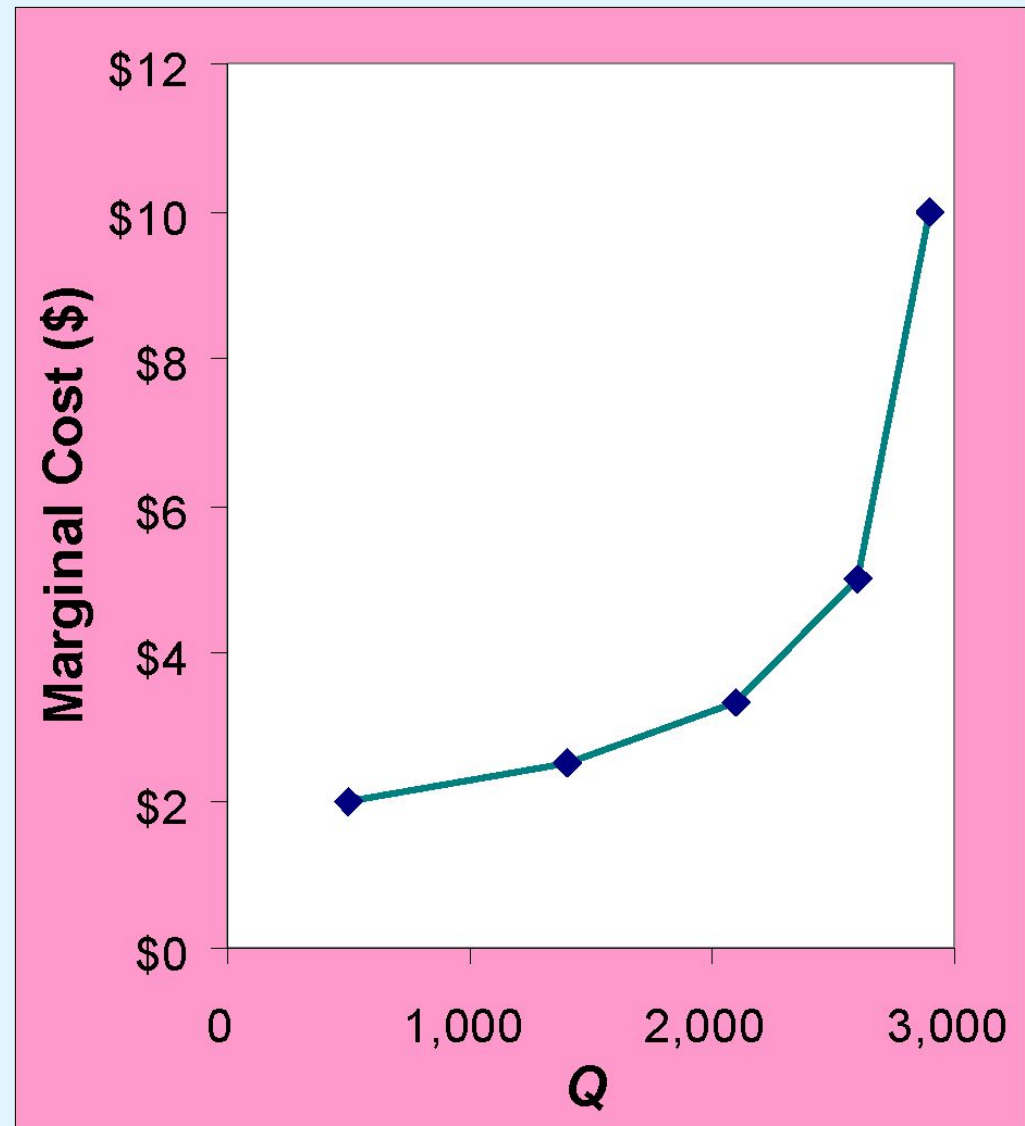
- Marginal cost, MC
 - Increase in total cost arising from an extra unit of production
 - Marginal cost = Change in total cost / Change in quantity
 - $MC = \Delta TC / \Delta Q$
 - Increase in total cost
 - From producing an additional unit of output

EXAMPLE 1: Total and Marginal Cost

	Q (bushels of wheat)	Total Cost	Marginal Cost (MC)
	0	\$1,000	
$\Delta Q = 1000$	1000	\$3,000	\$2.00
$\Delta Q = 800$	1800	\$5,000	\$2.50
$\Delta Q = 600$	2400	\$7,000	\$3.33
$\Delta Q = 400$	2800	\$9,000	\$5.00
$\Delta Q = 200$	3000	\$11,000	\$10.00

EXAMPLE 1: The Marginal Cost Curve

Q (bushels of wheat)	TC	MC
0	\$1,000	
		\$2.00
1000	\$3,000	
		\$2.50
1800	\$5,000	
		\$3.33
2400	\$7,000	
		\$5.00
2800	\$9,000	
		\$10.00
3000	\$11,000	





Why MC Is Important

- Farmer Jack is rational and wants to maximize his profit
 - To increase profit, should he produce more or less wheat?
 - Farmer Jack needs to “think at the margin”
 - If the cost of additional wheat (MC) is less than the revenue he would get from selling it, then Jack’s profits rise if he produces more.



Fixed and Variable Costs

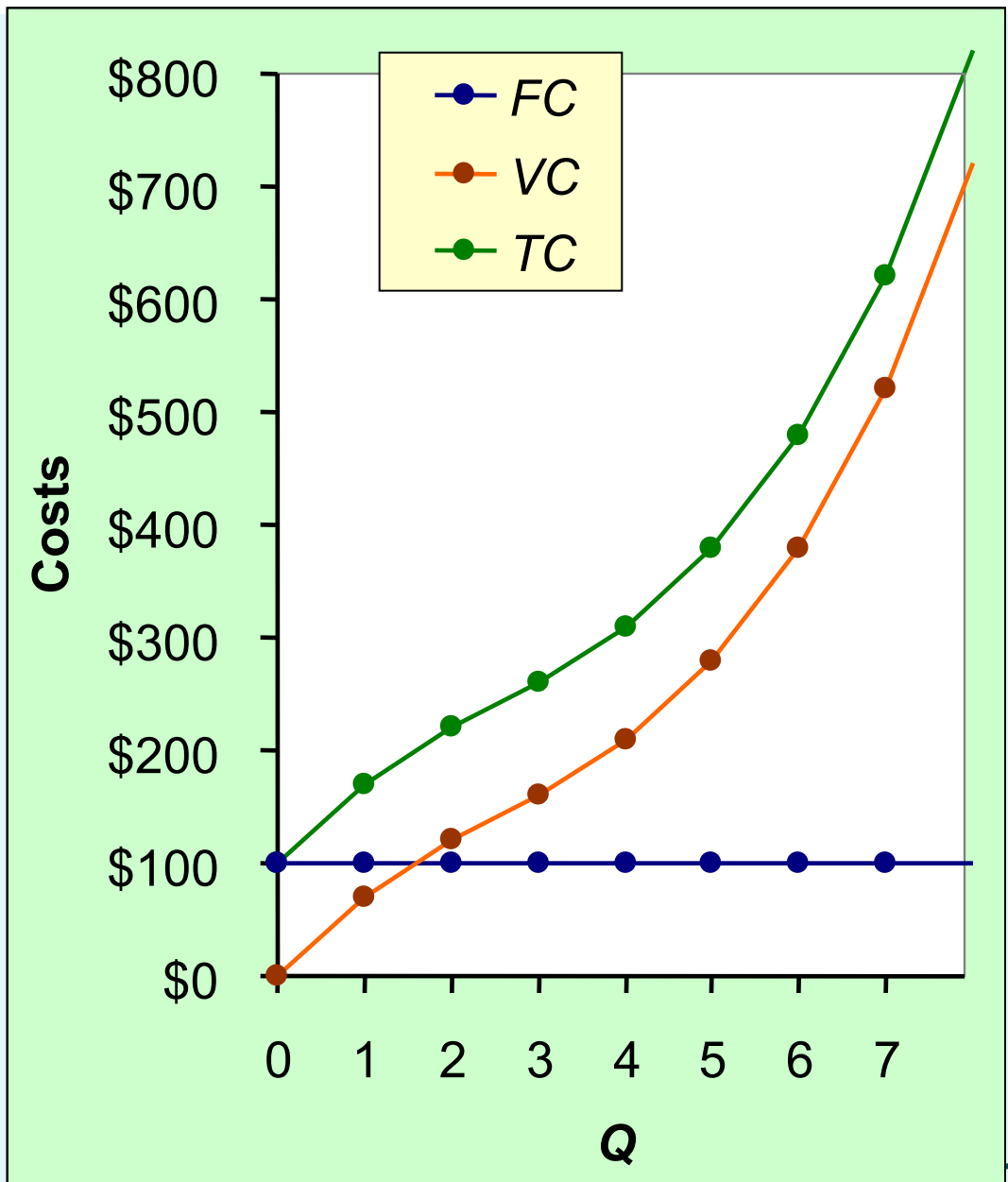
- **Fixed costs, FC**, do not vary with the quantity of output produced
 - For Farmer Jack, $FC = \$1000$ for his land
 - Other examples: cost of equipment, loan payments, rent
- **Variable costs, VC**, vary with the quantity of output produced
 - For Farmer Jack, $VC =$ wages he pays workers
 - Other example: cost of materials
- **Total cost = Fixed cost + Variable cost**

EXAMPLE 2: Production Costs

- Our second example is more general, applies to any type of firm producing any good with any types of inputs.
 - Calculate and graph TC knowing FC and VC
 - Calculate and graph marginal and average costs
 - Understand the relationship between marginal cost and average cost

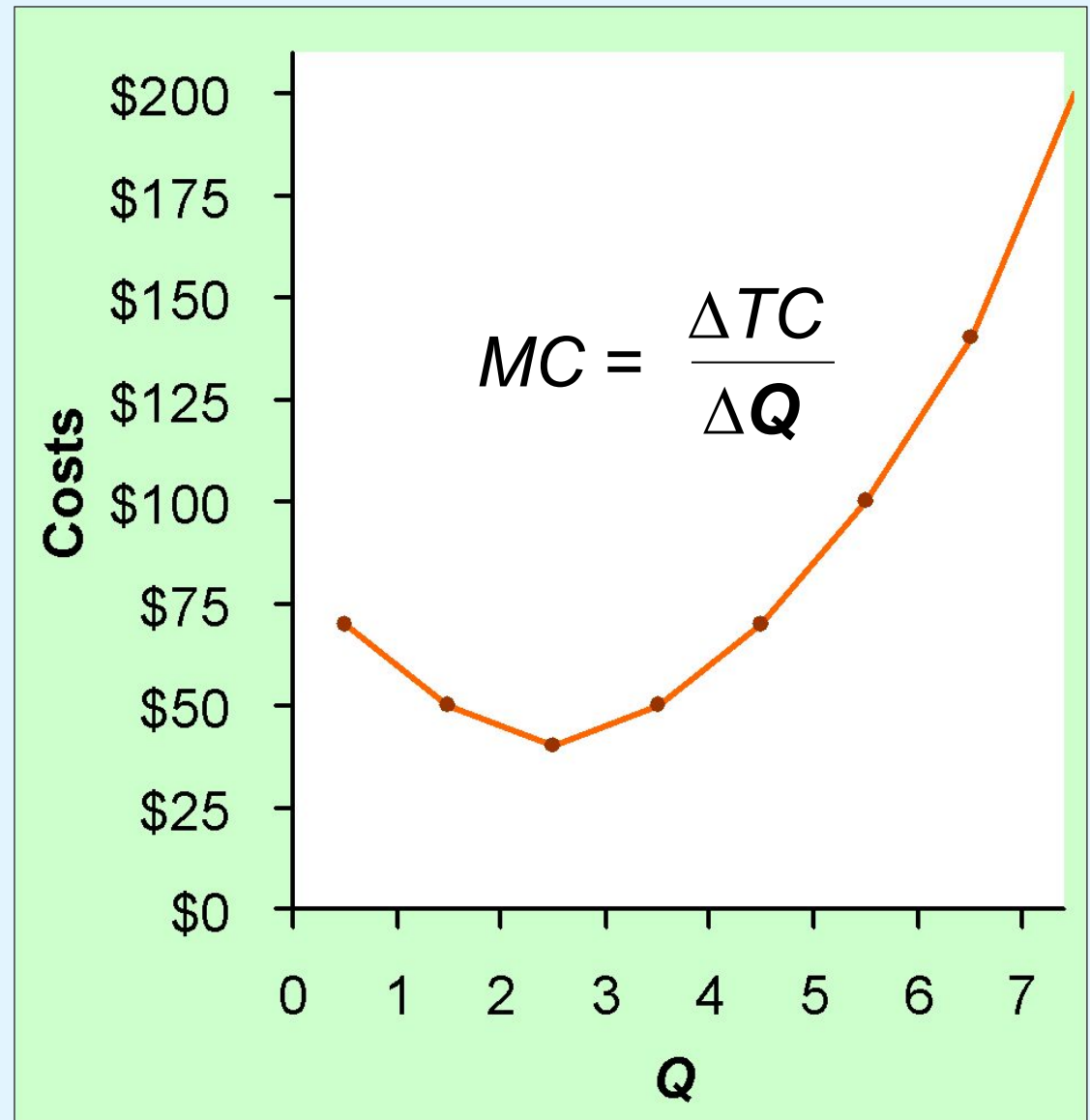
EXAMPLE 2: Costs: $TC = FC + VC$

Q	FC	VC	TC
0	\$100	\$0	\$100
1	100	70	170
2	100	120	220
3	100	160	260
4	100	210	310
5	100	280	380
6	100	380	480
7	100	520	620

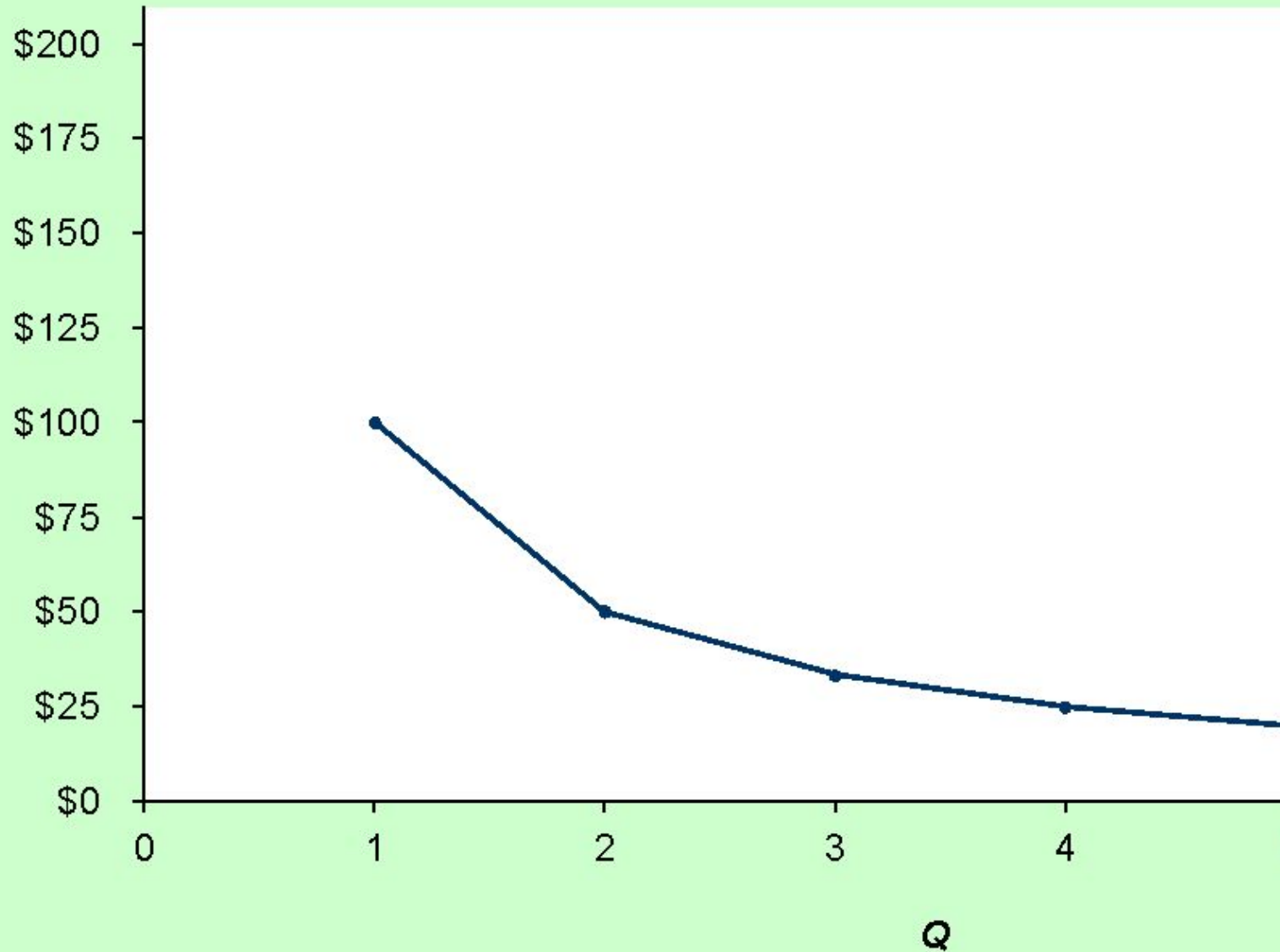


EXAMPLE 2: Marginal Cost

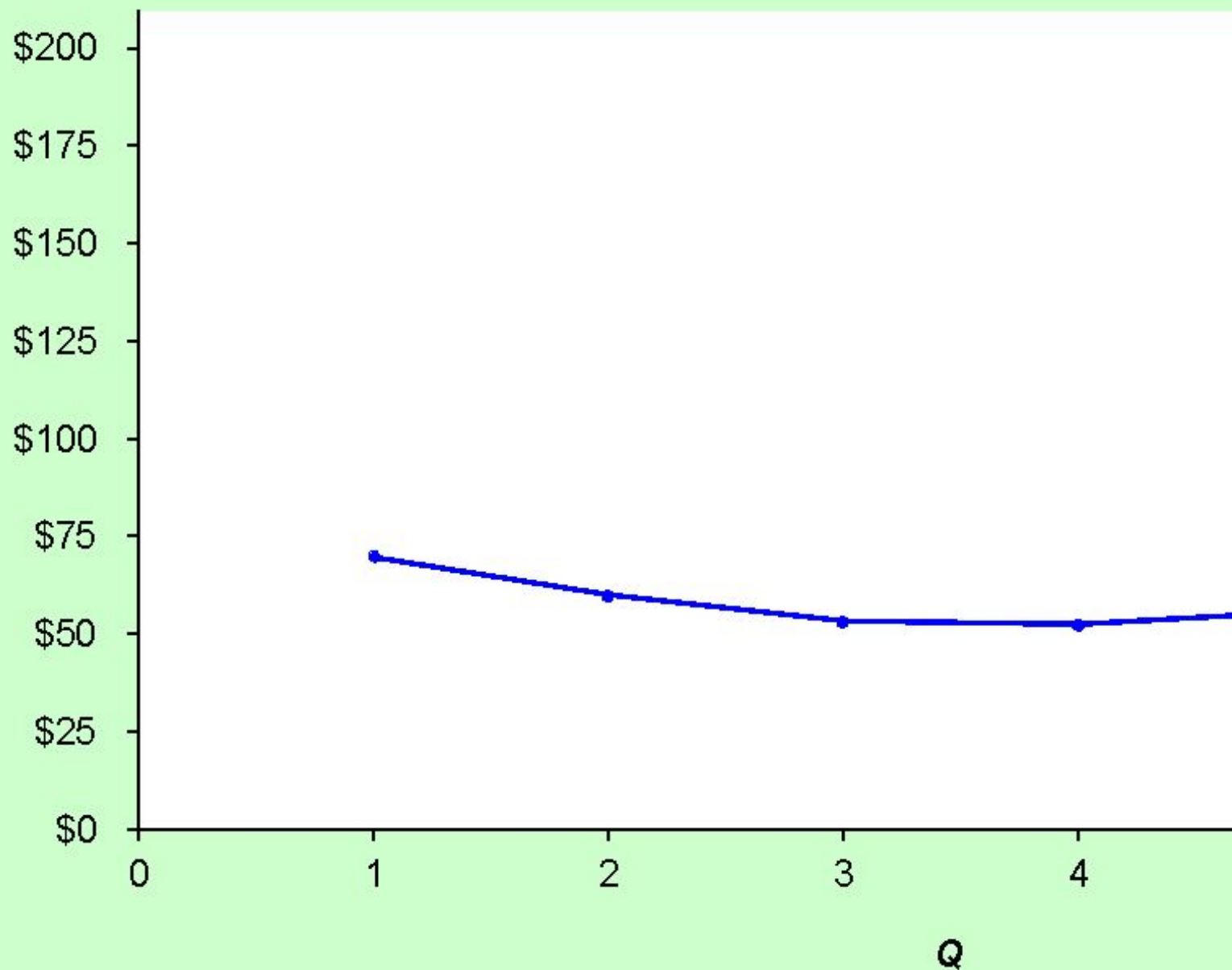
<i>Q</i>	<i>TC</i>	<i>MC</i>
0	\$100	
1	170	\$70
2	220	50
3	260	40
4	310	50
5	380	70
6	480	100
7	620	140



EXAMPLE 2: Average Fixed Cost, AFC



EXAMPLE 2: Average Variable Cost ΔVC



EXAMPLE 2: Average Total Cost

<i>Q</i>	<i>TC</i>	<i>ATC</i>	<i>AFC</i>	<i>AVC</i>
0	\$100	n/a	n/a	n/a
1	170	\$170	\$100	\$70
2	220	110	50	60
3	260	86.67	33.33	53.33
4	310	77.50	25	52.50
5	380	76	20	56.00
6	480	80	16.67	63.33
7	620	88.57	14.29	74.29

Average total cost (ATC) equals total cost divided by the quantity of output:

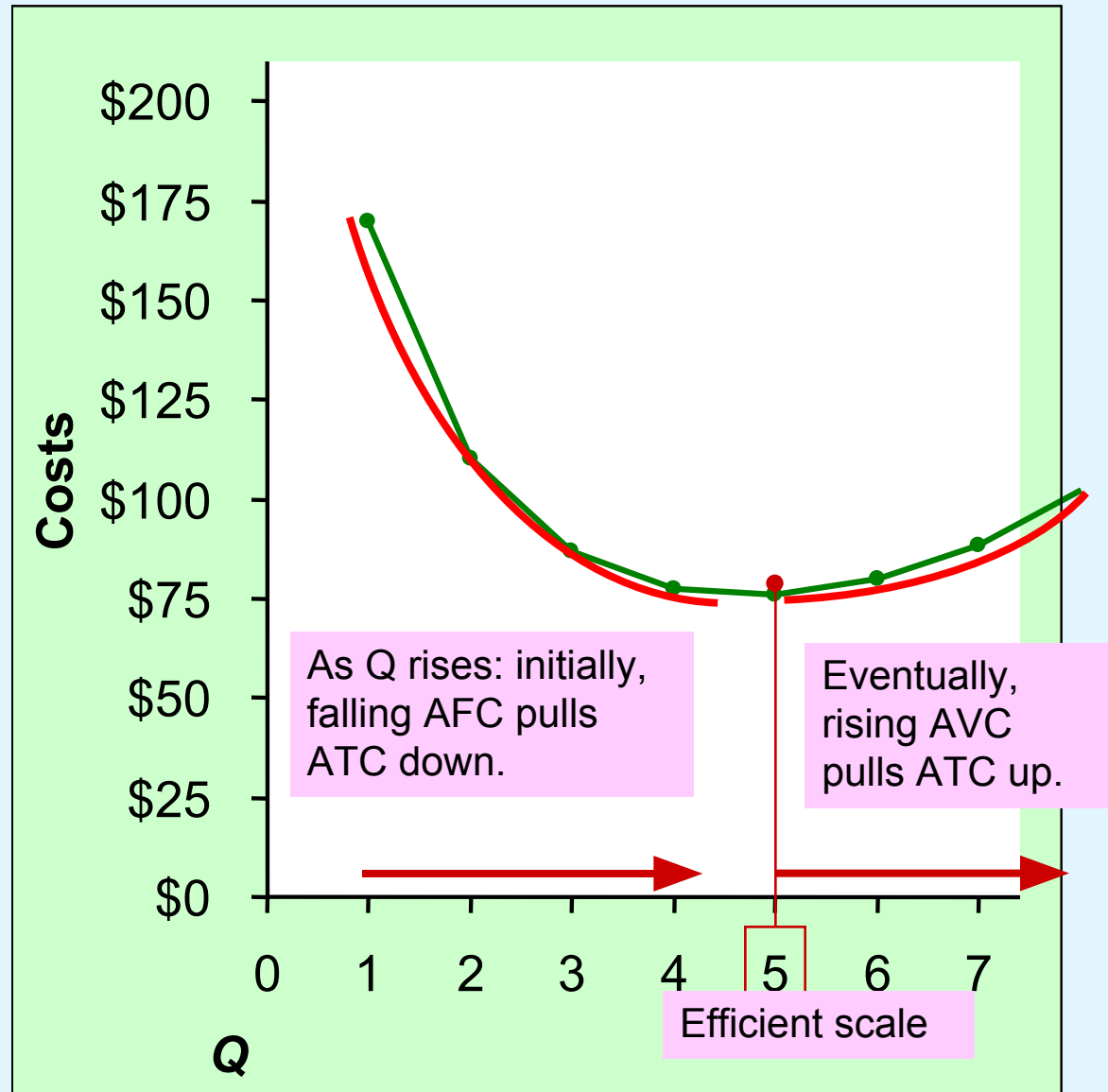
$$ATC = TC/Q$$

Also,

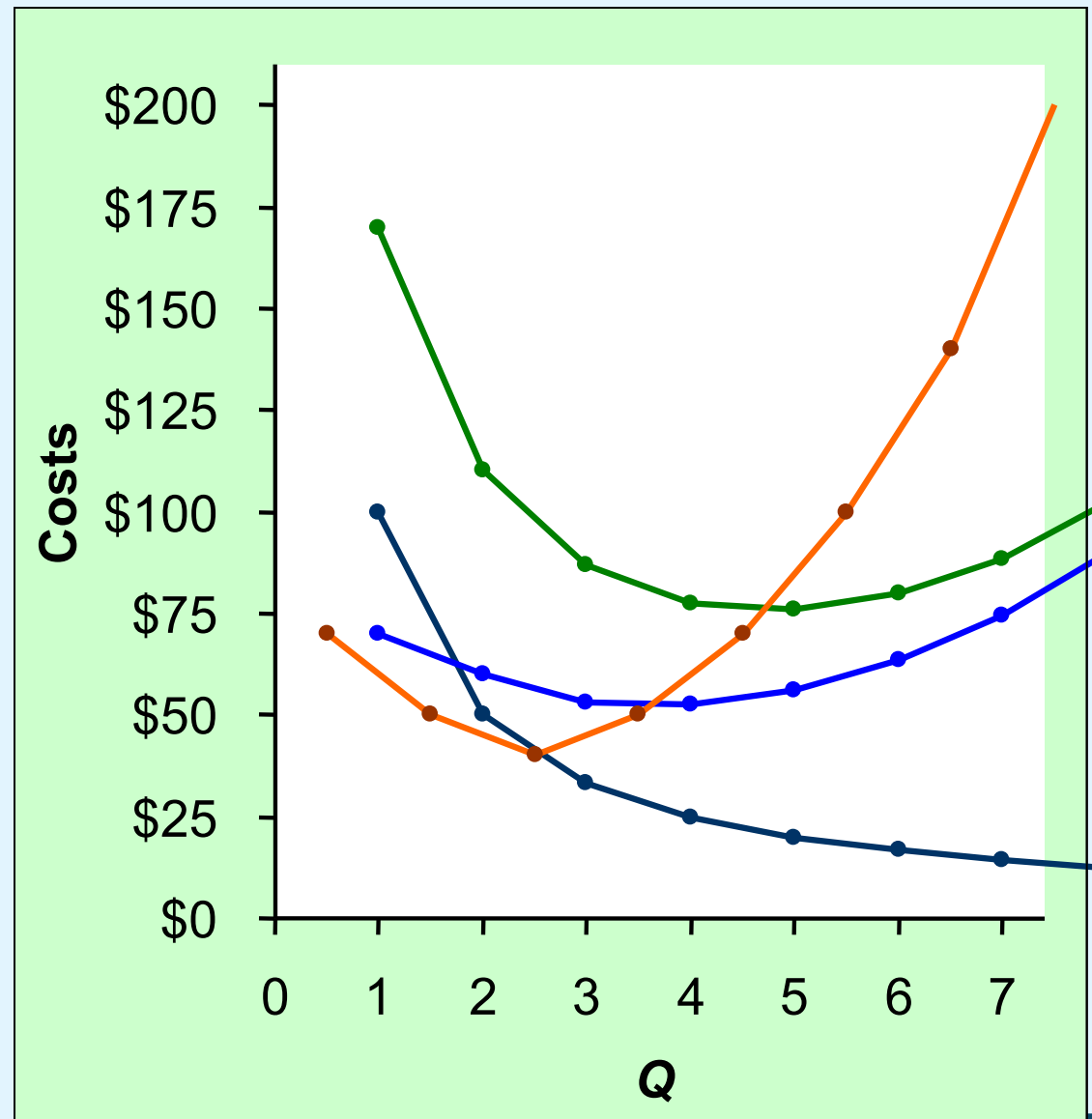
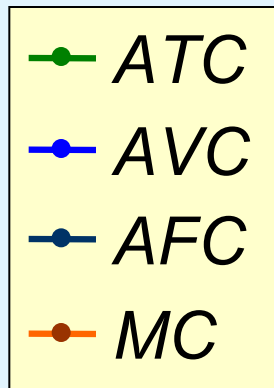
$$ATC = AFC + AVC$$

EXAMPLE 2: Average Total Cost, usually U-shaped

Q	TC	ATC
0	\$100	n/a
1	170	\$170
2	220	110
3	260	86.67
4	310	77.50
5	380	76
6	480	80
7	620	88.57



EXAMPLE 2: The Various Cost Curves Together

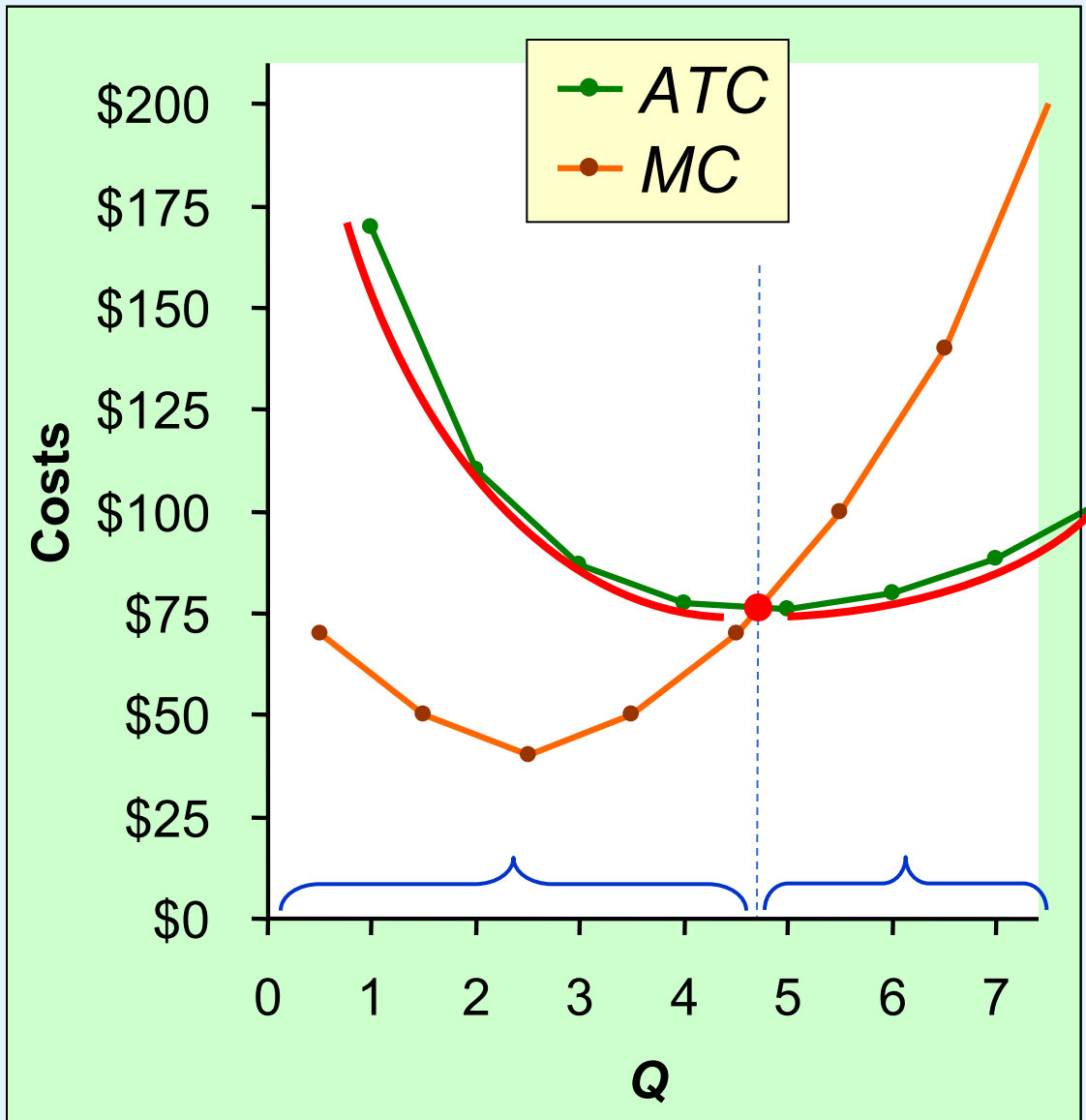


EXAMPLE 2: ATC and MC

When $MC < ATC$,
ATC is falling.

When $MC > ATC$,
ATC is rising.

The MC curve
crosses the ATC
curve
at the ATC curve's
minimum.



Fill in the blank spaces of this table.

Q	VC	TC	AFC	AVC	ATC	MC
0		\$50	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	\$10
1	10			\$10	\$60.00	
2	30	80				
3			16.67	20	36.67	30
4	100	150	12.50		37.50	
5	150			30		
6	210	260	8.33	35	43.33	60

Active Learning 3

Answers

First, deduce $FC = \$50$ and use $FC + VC = TC$.

Q	VC	TC	AFC	AVC	ATC	MC
0	\$0	\$50	n/a	n/a	n/a	\$10
1	10	60	\$50.00	\$10	\$60.00	
2	30	80	25.00	15	40.00	20
3	60	110	16.67	20	36.67	30
4	100	150	12.50	25	37.50	40
5	150	200	10.00	30	40.00	50
6	210	260	8.33	35	43.33	60



Costs in the Short Run & Long Run

- Short run:
 - Some inputs are fixed (e.g., factories, land)
 - The costs of these inputs are FC
- Long run:
 - All inputs are variable (e.g., firms can build more factories or sell existing ones)
- In the long run
 - ATC at any Q is cost per unit using the most efficient mix of inputs for that Q (e.g., the factory size with the lowest ATC)

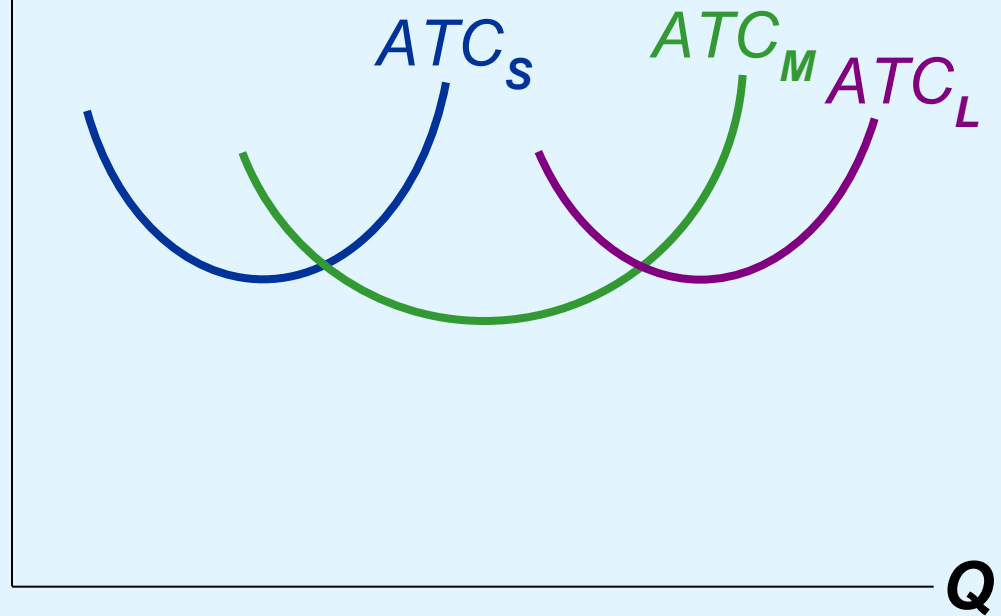
EXAMPLE 3: LRATC with 3 factory sizes

Firm can choose from three factory sizes: S, M, L.

Each size has its own SRATC curve.

The firm can change to a different factory size in the long run, but not in the short run.

Avg
Total
Cost

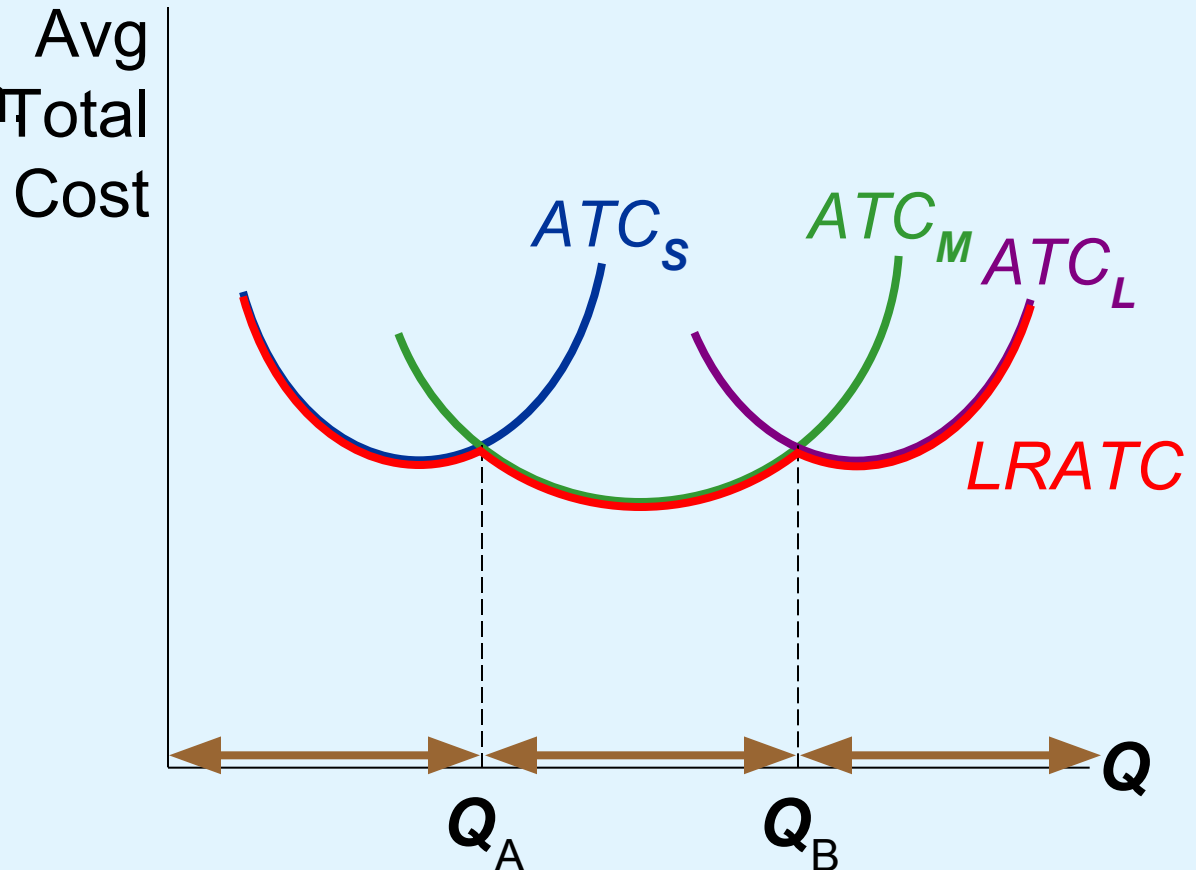


EXAMPLE 3: LRATC with 3 factory sizes

To produce less than Q_A , firm will choose size S in the long run

To produce between Q_A and Q_B , firm will choose size M in the long run.

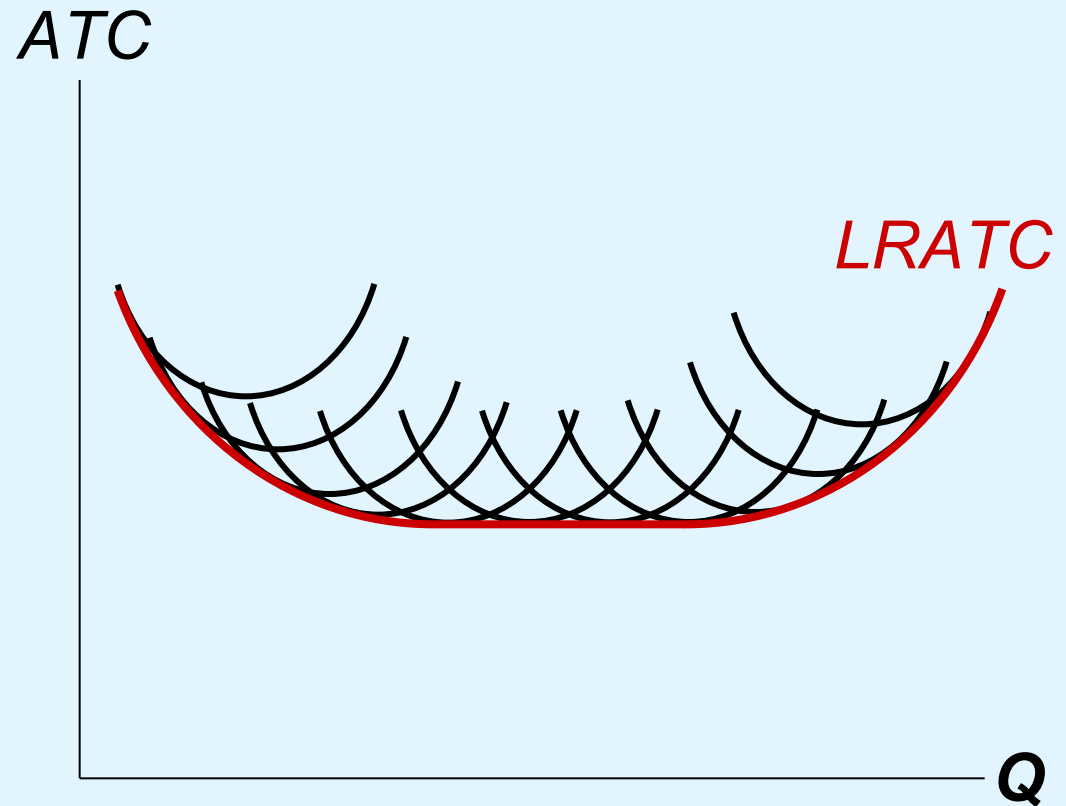
To produce more than Q_B , firm will choose size L in the long run.



A Typical LRATC Curve

In the real world, factories come in many sizes, each with its own SRATC curve.

So a typical LRATC curve looks like this:



How ATC Changes as the Scale of Production Changes

Economies of scale:

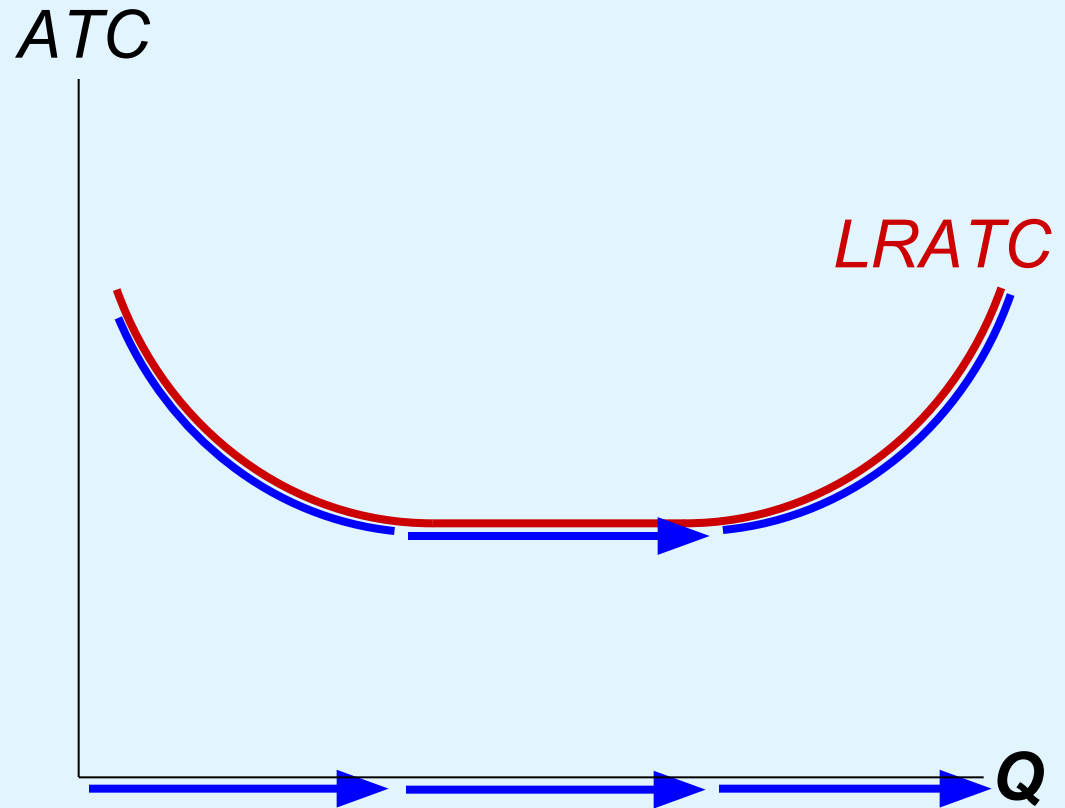
ATC falls as Q increases.

Constant returns to scale:

ATC stays the same as Q increases.

Diseconomies of scale:

ATC rises as Q increases.





Costs in Short and Long Run

- Economies of scale
 - Long-run average total cost falls as the quantity of output increases
 - Increasing specialization among workers
 - More common when Q is low
- Constant returns to scale
 - Long-run average total cost stays the same as the quantity of output changes



Costs in Short and Long Run

- Diseconomies of scale
 - Long-run average total cost rises as the quantity of output increases
 - Increasing coordination problems in large organizations.
 - E.g., management becomes stretched, can't control costs.
 - More common when Q is high.

Summary

- The goal of firms is to maximize profit, which equals total revenue minus total cost.
- When analyzing a firm's behavior, it is important to include all the opportunity costs of production.
 - Explicit: wages a firm pays its workers
 - Implicit: wages the firm owner gives up by working at the firm rather than taking another job
- Economic profit takes both explicit and implicit costs into account, whereas accounting profit considers only explicit costs.

Summary

- A firm's costs reflect its production process.
 - Diminishing marginal product: production function gets flatter as Q of an input increases
 - Total-cost curve gets steeper as the quantity produced rises.
- Firm's total costs = fixed costs + variable costs.
 - Fixed costs: do not change when the firm alters the quantity of output produced.
 - Variable costs: change when the firm alters the quantity of output produced.

Summary

- Average total cost is total cost divided by the quantity of output.
- Marginal cost is the amount by which total cost rises if output increases by 1 unit.
- Graph average total cost and marginal cost.
 - Marginal cost rises with the quantity of output.
 - Average total cost first falls as output increases and then rises as output increases further.
 - The marginal-cost curve always crosses the average total-cost curve at the minimum of average total cost

Summary

- A firm's costs often depend on the time horizon considered.
 - In particular, many costs are fixed in the short run but variable in the long run.
 - As a result, when the firm changes its level of production, average total cost may rise more in the short run than in the long run.